

Parking Allocation Platform

Full Product Requirements & Design Blueprint

User Roles · Booking Flows · Payments · Access Control
Notifications · Analytics · Integrations · Build Phases

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1. User Roles

Role	Description	Key Capabilities
Customer	End users who park their vehicles	Register vehicles · Search slots · Book & pay · View history · Cancel & extend · Rate spaces
Parking Owner	Businesses or individuals who list parking spaces	Register spaces · Manage slots & pricing · View bookings · Track earnings · Set availability
Admin	Platform operators who oversee the marketplace	Verify owners & spaces · Handle disputes · Manage fees · View analytics · Issue refunds
System (Automated)	Automated processes that run without human input	Real-time slot tracking · Auto-charge overstay · Send reminders · Fraud detection · Payout processing

2. Customer Booking Flow

01 Register & Login

Provide name, email, phone, password. Verify phone via OTP. Register vehicle: plate number, vehicle type (2W/4W/EV/heavy).

02 Search for a Slot

Enter location or enable GPS. Filter by: date & time range, vehicle type, covered/open, EV charging, CCTV, 24/7, price range. View results on map or list view with real-time slot counts.

03 Select a Parking Space

View space details: photos, amenities, address, entry instructions, cancellation policy, owner rating. Select a specific slot (if exposed) or let system auto-assign.

04 Confirm Booking

Review booking summary: space name, slot, date/time, duration, subtotal, taxes, total. Apply promo/coupon code if available.

05 Payment

Authorize payment (hold). Supported: Credit/Debit card, UPI, Net Banking, Wallet. Payment confirmation required before booking is confirmed.

06 Booking Confirmed

Receive QR code ticket via SMS + email. Booking visible in 'My Bookings' dashboard. Owner & system notified.

07 Entry

Arrive at parking space. Show QR at gate or ANPR auto-reads plate. Entry timestamp logged. Slot marked 'occupied' in real-time.

08 Exit & Settlement

Exit scan records departure time. If within booked window: payment already captured. If overstay: additional charge auto-deducted. Receipt sent.

3. Parking Owner Onboarding Flow

01 Register Account

Business name, owner name, email, phone, OTP verification.

02 KYC & Verification

Upload: Govt ID proof, Address proof, Ownership/lease document, GST certificate (if applicable). Admin reviews within 24–48 hours.

03 Add Parking Space

Address with GPS pin, photos (min 3), total capacity, space type (open/covered/multi-level), operating hours, entry/exit instructions.

04 Define Slot Layout

Add slot numbers/names, assign vehicle types per slot (2W, 4W, EV, handicap), mark special slots.

05 Set Pricing

Hourly rate, daily rate, monthly pass rate, peak hour surge multiplier, weekend rates, free cancellation window.

06 Admin Verification

Admin verifies documents and space details. May request corrections or additional photos.

07 Go Live

Space appears in search results. Owner receives dashboard access: bookings, earnings, calendar view.

08 Payout Setup

Add bank account / UPI for automated payouts. Platform deducts commission (e.g. 15%) and transfers balance weekly/monthly.

4. Core Platform Modules

■ Search & Discovery

- Map view with live slot availability (colour-coded: green=free, red=full, amber=few left)
- Filters: covered/open, EV charging, CCTV surveillance, 24/7 access, handicap accessible
- Sort by: distance, price (low-high), rating, availability
- Real-time 'X slots remaining' counter per space
- Saved/favourite spaces for quick re-booking

■ Booking Management

- Duration types: hourly, daily, weekly, monthly subscription
- Advance booking up to 30 days
- Extend active booking via app (subject to availability)
- Cancellation with configurable refund tiers
- Waitlist: auto-notify customer when slot becomes available
- Booking status: Pending → Confirmed → Active → Completed / Cancelled

■ Vehicle Management

- Register multiple vehicles per account
- Fields: plate number, vehicle type (2W/4W/EV/Heavy), make, model, colour
- Default vehicle selection for quick booking
- Plate number used for ANPR matching at gate

■ Review & Rating System

- Customer rates parking space (1–5 stars) after checkout
- Comments: cleanliness, security, ease of entry/exit
- Owner can respond to reviews
- Minimum 5 reviews before space shows aggregate score
- Admin can remove abusive or fake reviews

■ Customer Dashboard

- Upcoming bookings with QR codes
- Active booking with time remaining + extend button
- Booking history with receipts/invoices
- Saved spaces & vehicles
- Wallet balance & transaction history
- Dispute / complaint submission

■ Owner Dashboard

- Calendar view of all bookings per space
- Real-time occupancy heatmap
- Revenue summary: daily/weekly/monthly
- Pending payouts and payout history
- Slot management: block/unblock, add/remove
- Customer feedback & ratings received

5. Payments & Billing

Event	Action	Notes
Booking confirmed	Authorization hold on card/wallet	Amount held = booking total + overstay buffer
Customer checks in	Hold converted to capture	Timestamp-linked to entry scan
Customer checks out	Final charge calculated	Overstay delta added if applicable
Cancellation (early)	Full refund released	>2 hrs before booking start
Cancellation (late)	Partial refund per policy	Configurable: e.g. 50% within 2 hrs
No-show	Cancellation fee charged	After grace period (default: 15 min)
Overstay	Overstay rate charged on exit	Rate = 1.5x hourly rate
Owner payout	Net amount transferred to bank/UPI	Weekly; platform fee deducted first
Refund (dispute)	Admin-initiated full/partial refund	Settled within 5–7 business days

Supported Payment Methods: Credit/Debit Card (Visa, Mastercard, RuPay) · UPI (GPay, PhonePe, Paytm) · Net Banking · Platform Wallet · Corporate Invoice

Billing Documents: GST-compliant invoices auto-generated · Downloadable PDF receipts · Monthly statements for owners

6. Access Control

- **QR Code Tickets:** Unique per booking, expire after use, regenerate if lost via app. One-time scan only.
- **ANPR (Phase 2):** Camera reads plate number on entry/exit. System auto-matches to active booking. No human intervention needed.
- **Entry Logging:** Timestamp, slot ID, booking ID, vehicle plate logged on every entry and exit.
- **Gate / Barrier API:** Webhook triggers gate open on valid QR scan or ANPR match. Supports boom barriers, bollards, sliding gates.
- **Manual Override:** Security staff can manually verify and open gate via admin app (for camera failures or QR scan failures).
- **Multi-entry Pass:** Monthly subscribers get unlimited entries within booking window.

7. Notifications

Trigger	Channel	Recipient
Booking confirmed	SMS + Email + Push	Customer
Payment receipt	Email	Customer
Booking reminder (30 min before)	SMS + Push	Customer
Booking expiry warning (15 min)	Push	Customer
Check-out confirmation	SMS + Email	Customer
Overstay charge applied	Push + Email	Customer
Cancellation confirmed	SMS + Email	Customer
Waitlist slot available	SMS + Push	Customer

New booking received	Push + Email	Owner
Booking cancelled	Push + Email	Owner
Payout processed	Email	Owner
New dispute raised	Email	Owner + Admin
Space verification approved	Email	Owner
Fraud alert	Email + SMS	Admin

8. Analytics & Reporting

Owner Analytics

- Daily / weekly / monthly revenue
- Occupancy rate % per slot
- Peak hours heatmap (hour x day grid)
- Top revenue-generating slots
- Cancellation & no-show rate
- Customer review trends
- Payout history & projections

Admin Analytics

- Platform-wide GMV (Gross Merchandise Value)
- Active users (DAU / MAU)
- Booking conversion funnel
- Top cities / zones by revenue
- Owner acquisition & churn
- Dispute rate & resolution time
- Payment failure rate by gateway

9. Admin Panel

- Owner management: view pending KYC submissions, approve/reject/suspend owners
- Space management: verify space listings, flag suspicious listings, force-unpublish
- Booking oversight: view all bookings, manually resolve entry/exit discrepancies
- Dispute management: receive dispute tickets, review evidence (photos, logs), issue refunds
- Platform fee configuration: set commission %, overstay rates, cancellation policies globally
- User management: ban/suspend accounts, reset passwords, view activity logs
- Promo & coupon management: create discount codes, usage limits, expiry dates
- System health: monitor booking API latency, payment gateway uptime, error rates
- Content moderation: review flagged reviews, respond to owner/customer complaints

10. Non-Functional Requirements

Category	Requirement	Target
Availability	Platform uptime SLA	99.9% (< 9 hrs downtime/year)
Performance	Slot availability API response time	< 200ms (p95)
Performance	Booking confirmation end-to-end	< 3 seconds
Scalability	Concurrent active bookings	Support 100K+ simultaneously
Concurrency	Race condition on slot booking	Row-level DB locks + idempotency keys
Security	Payment data handling	PCI-DSS Level 1 compliant
Security	User data at rest	AES-256 encryption
Security	Data in transit	TLS 1.3 enforced
Privacy	User data regulation	DPDP Act (India) / GDPR compliant
Mobile	Platform support	iOS 14+ and Android 9+ native apps
Accessibility	Web UI standard	WCAG 2.1 AA
Reliability	Payment transaction integrity	Idempotent, atomic, rollback-safe
Rate Limiting	Booking API abuse prevention	Max 10 booking attempts/min per user
Disaster Recovery	RTO / RPO	RTO < 1hr, RPO < 15 min

11. Third-Party Integrations

Category	Options	Purpose
Payment Gateway	Razorpay / Stripe / PayU	Card, UPI, Wallet, Net Banking processing
Maps & Geocoding	Google Maps API / Mapbox	Search UI, space pins, distance calculation
SMS Gateway	Twilio / MSG91 / Exotel	OTP, booking confirmations, reminders
Email Service	SendGrid / Amazon SES	Transactional emails, invoices
Push Notifications	Firebase FCM / APNs	Real-time alerts on iOS & Android
KYC Verification	Aadhaar API / DigiLocker / Signzy	Owner identity & document verification
ANPR Camera	Custom / Genetec / VIVOTEK (Phase 2)	Automated plate recognition at gate
IoT Gate Control	Custom REST API / MQTT	Boom barrier / door controller triggers
Accounting Export	CSV / Tally / QuickBooks API	Owner bookkeeping & tax filing
Analytics	Mixpanel / Amplitude / custom BI	User behaviour & funnel analysis
Support Chat	Freshdesk / Intercom / Zendesk	Customer support ticketing
Fraud Detection	Sift / custom ML rules	Anomalous payment pattern detection

12. Edge Cases & Business Rules

Booking Conflicts

- Two users book same slot simultaneously → DB row-level locking + optimistic concurrency; second request gets 'slot unavailable' error
- Owner marks slot unavailable after customer books → customer notified immediately, full auto-refund within 24 hrs, slot unlocked
- Payment succeeds but booking fails (network timeout) → idempotency key prevents double charge; booking retried or refunded automatically
- Customer tries to book a slot they already have an active booking in → system blocks duplicate and prompts to extend

Overstay & No-Show

- Customer overstays booked window → on exit scan, system charges (overstay hours × overstay rate). Rate = 1.5× hourly rate by default (configurable per owner)
- Customer does not arrive → slot auto-released after grace period (default 15 min). Booking marked 'No-show'. No-show fee charged per policy
- Entry scan fails (QR lost or camera down) → security staff uses admin app to manually verify and log entry
- Customer extends booking while parked → system checks next booking on same slot; extension allowed only if no conflict

Cancellations & Refunds

- Cancel > 2 hours before start → 100% refund (configurable threshold per owner)
- Cancel < 2 hours before start → partial refund (e.g. 50% after platform fee, configurable)
- Cancel after entry (slot activated) → no refund for elapsed time; remaining pro-rated may be refunded per owner policy
- Owner cancels confirmed booking → 100% refund to customer + penalty deducted from owner's payout
- Dispute raised → admin mediation workflow; resolution within 5 business days; auto-refund if unresolved

System & Availability

- Space is full but customer tries to book → waitlist offered; customer notified when slot opens (FIFO queue)
- Payment gateway downtime → queue payment attempt, notify user, retry automatically up to 3 times
- Space photos fail to load → placeholder shown; booking not blocked
- GPS unavailable → user can manually enter address for search

13. Security Requirements

Area	Requirement
Authentication	JWT-based auth with refresh tokens. OAuth2 SSO (Google/Apple sign-in). OTP for phone verification and sensitive operations (password reset, payout change).
Authorisation	Role-based access control (RBAC): Customer / Owner / Admin roles with strict permission scopes. Owners can only access their own spaces and bookings. Customers can only access their own data.
QR Code Security	QR codes are: time-limited (expire if unused after booking window), single-use (invalidated after first valid scan), signed with HMAC to prevent forgery.
API Security	Rate limiting on all public endpoints. CAPTCHA on registration and login. IP-based anomaly detection. WAF (Web Application Firewall) in front of all APIs.
Data Privacy	PII (names, plates, payment tokens) encrypted at rest (AES-256). Sensitive fields masked in logs. Employees access user data only on need-to-know basis with audit logs.
Payment Security	No raw card data stored — tokenized by payment gateway (PCI-DSS). 3D Secure (OTP confirmation) for card payments. Card CVV never stored.
Audit Logs	All admin actions logged with timestamp, actor, IP, and action type. Immutable audit trail for bookings, payments, and access events.
Fraud Prevention	Flag: multiple failed payments from same device, high-velocity bookings from same account, ANPR mismatch (plate at gate ≠ plate on booking).

14. Build Phases & Priority

Phase 1 — MVP (0–3 months)

- User & owner registration, login, OTP verification
- Vehicle registration (plate, type)
- Parking space listing (manual admin verification)
- Slot search — list view with basic filters
- Booking flow with payment (card + UPI via Razorpay/Stripe)
- QR code generation and manual entry verification
- Email + SMS confirmation notifications
- Basic owner dashboard (bookings list, revenue total)
- Admin panel: approve owners, view all bookings
- Cancellation & refund (manual trigger)

Phase 2 — Growth (3–6 months)

- Map view with real-time slot availability
- Extend booking while parked
- Waitlist feature with auto-notification
- Overstay auto-detection and charge on exit
- Owner analytics dashboard (occupancy, revenue charts)
- Review & rating system
- Monthly subscription / pass
- Platform wallet & credits
- Promo & coupon codes
- Automated payout processing with commission split
- Dispute management workflow for admin

Phase 3 — Scale (6–12 months)

- ANPR plate recognition at entry/exit
- IoT gate / barrier controller integration
- Dynamic surge pricing (peak hour multiplier)
- EV charging slot booking
- Corporate accounts & bulk booking
- AI occupancy prediction (fill rate forecasting)
- Multi-city / franchise management
- White-label SDK for malls, airports, campuses
- Advanced fraud detection (ML-based)
- Mobile app (iOS + Android native)

15. Tech Stack Recommendations

Layer	Recommended	Rationale
Mobile App	React Native / Flutter	Single codebase for iOS + Android; faster iteration
Web Frontend	Next.js (React)	SEO-friendly for discovery pages; SSR + CSR
Backend API	Node.js (Express) or Django	Node for high concurrency; Django for rapid CRUD
Real-time	WebSocket / Socket.io	Live slot availability updates without polling
Database	PostgreSQL (primary)	ACID compliance critical for booking & payments
Cache	Redis	Slot availability cache, session store, rate limiting
Queue	BullMQ / AWS SQS	Async jobs: notifications, payout processing
Object Storage	AWS S3 / Cloudflare R2	Space photos, KYC documents, invoices
Maps	Google Maps Platform	Geocoding, directions, nearby search
Infrastructure	AWS / GCP	Auto-scaling, managed RDS, CDN (CloudFront)
Auth	JWT + Firebase Auth / Auth0	OTP, OAuth2 (Google/Apple), token management
CI/CD	GitHub Actions	Automated testing, deployment pipelines
Monitoring	Datadog / Sentry	Error tracking, performance monitoring, alerts

This document covers the complete product requirements for a parking allocation platform. Scope and features should be validated with stakeholders before development begins. All third-party integrations are subject to vendor terms and pricing.